

Two-Axis Robotic Arm

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Top-Level Architecture Block Diagram

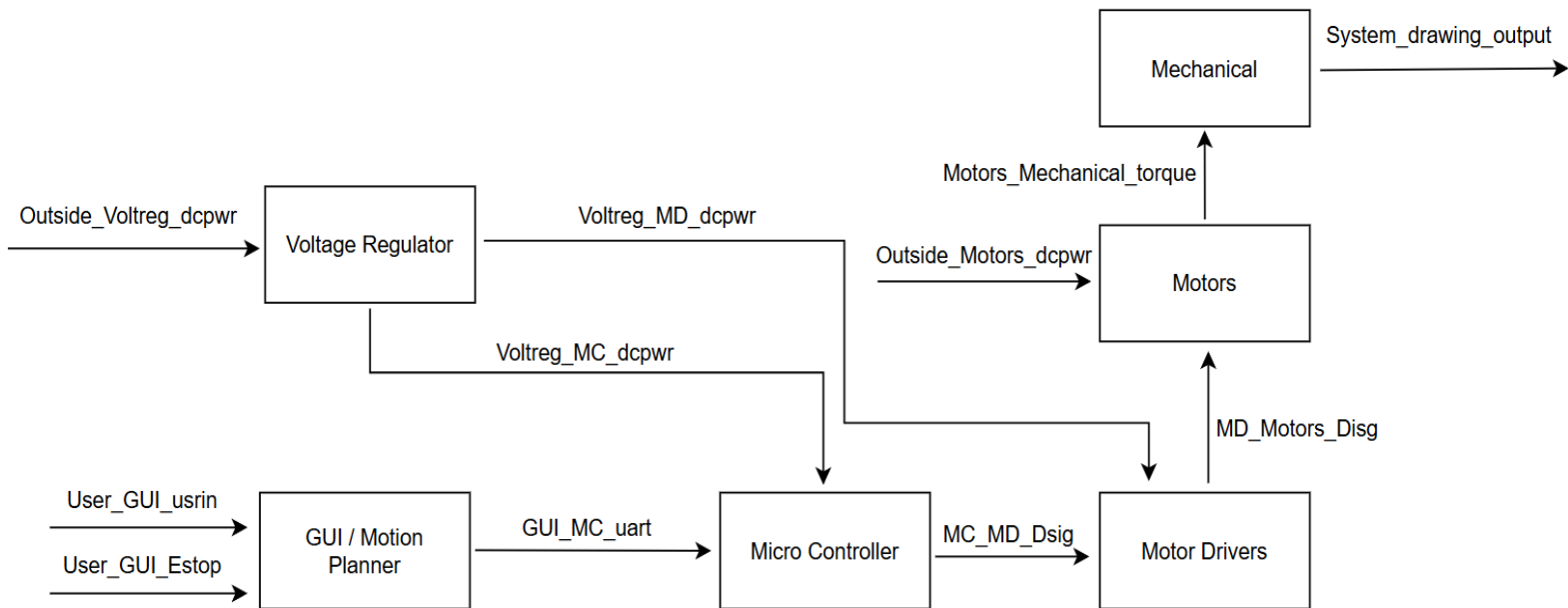


Fig 1. Top-Level Architecture Block Diagram.[1]

Block 1 Video Link

https://media.oregonstate.edu/media/t/1_diliv5mp

Block 1 Description

This power unit connects to an external adapter. The external adapter regulates the voltage supplied by a 120V outlet to provide a fixed 12V output with a maximum current of 3A [1]; subsequently, a Buck regulator converts this into a 5V output—also with a maximum current of 3A—to power the MCU [7]. Furthermore, this system is capable of supplying a 12V, 2A output to the motor drivers used for motor control under maximum load conditions [5] [6]. For circuit protection, a bulk capacitor has been placed at the power input stage to stabilize the power supply [3].

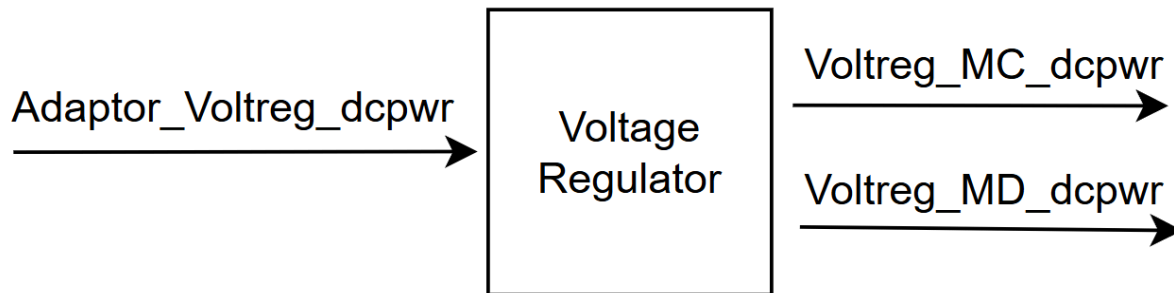


Fig. 2. Block-Level block diagram of Voltage Regulator.[1]

Block 1 Design Details

This power block receives 12V DC power—up to a maximum of 3A—from an external adapter via a barrel connector [1]. The block serves as the power management unit for the entire system. Input power is applied through the Adaptor_Voltreg_dcpwr interface; after passing through the barrel connector, it first flows through an electrolytic capacitor for input protection. This capacitor functions to smooth out any ripple voltage that may originate from the external adapter [3]. Following this stage, a stable 12V DC power supply is effectively established.

The resulting 12V power is then branched into two separate paths within the system. The first path, designated Voltreg_MD_dcpwr, connects directly to two stepper motor drivers. Motor operation demands the highest current consumption within the system (drawing a peak current of up to 1A per motor) [6]; therefore, to ensure a smooth and consistent current supply even under maximum load conditions, this path is designed to deliver power directly to the motor drivers without passing through a separate voltage regulator.

The second path leads to an LM2576-5.0WT switching buck converter, which supplies power to the microcontroller. Internally, this chip steps down the 12V input voltage to 5V. This stepped-down 5V output voltage passes through a catch diode, a power inductor, and an output capacitor before being delivered to the microcontroller as a stable DC power source. The microcontroller's peak current consumption is approximately 225mA [11]; calculating the total peak current for the entire system, including this value, yields:

$$I_{\text{total_peak}} = 2A \text{ (Motors)} + 0.225A = 2.225A.$$

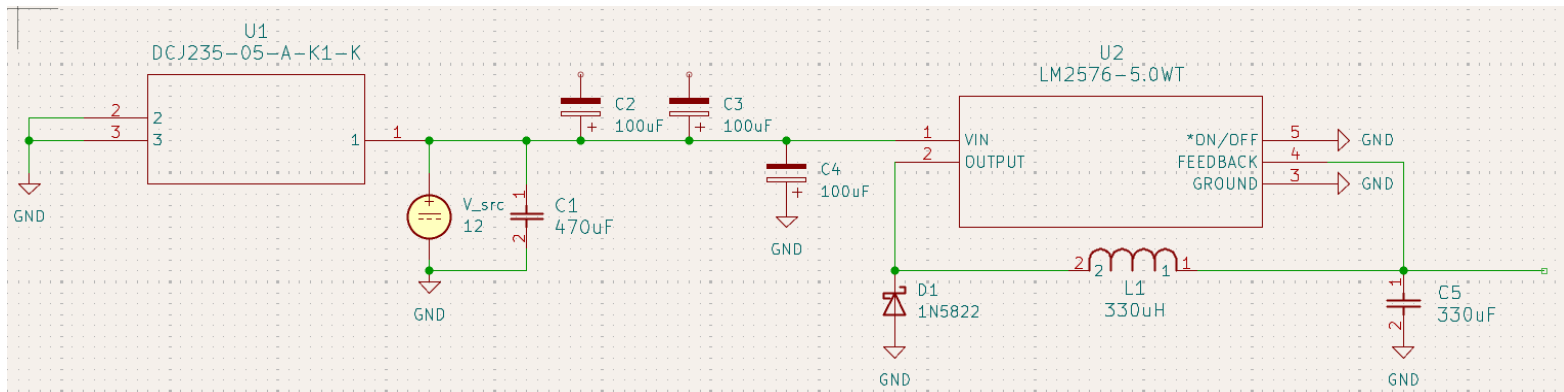


Fig 3. Schematic of Voltage Regulator.[1]

Block 1 Interface Validation

Interface property *Be sure to use the naming convention “from_to_type” here.	Why is this interface property this value?	How do you know your design details will meet or exceed this property? Cite your sources in IEEE.
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Adaptor_Voltreg_dcpwr: input

Vmax = 12.6V	A nominal voltage of 12 V is required to power two NEMA 17 stepper motors and a logic voltage regulator]. The maximum value of 12.6 V accounts for a potential +5% voltage fluctuation from an external adapter.	The adapter used in the system provides a 12V output. However, the components utilized in the system were selected for their high voltage ratings, enabling them to withstand voltages significantly exceeding 12V. For instance, the bulk capacitors are capable of withstanding a voltage of 25V [3].
Vmin = 11.4V	This value accounts for the voltage drop that may occur while the two motors are operating at their maximum capacity and drawing peak current.	According to the datasheet, this adapter has a voltage tolerance of 5% [1]. The LM2576 operates at a minimum voltage of 4V; therefore, 11.4V provides a sufficient operating margin [7].
Inom= 3.0A	The system's total peak current is approximately 2.15 A. In comparison, the nominal current rating of 3 A provides a margin of roughly 40%.	According to the datasheet, the power adapter is capable of delivering a maximum output of 3A [1]. Furthermore, the other components possess higher current ratings to ensure they can withstand a 3A output.

Voltreg_MD_dcpwr: Output

V _{max} = 12.6V	This represents the maximum voltage that can be applied to the motor driver. Since there is no voltage drop between the adapter and the motor driver, this value corresponds exactly to the adapter's maximum output voltage, taking into account the adapter's 5% tolerance.	Since the maximum operating voltage of the TMC2209 is 28V, 12.6V falls within the operating range [5].
V _{min} = 11.4V	This represents the maximum voltage drop that may temporarily occur when the motor is under maximum load. Similarly, since there is no voltage drop between the adapter and the motor driver, this corresponds exactly to the adapter's minimum input voltage minus a 5% drop.	The TMC2209 motor driver operates down to a minimum of 4.75V; therefore, 11.4V falls within its operating range [5].
I _{peak} = 1A	While the motor driver itself has a peak current (I _{peak}) of 2.8A, the maximum allowable current for the NEMA 17 motor used in the system is 1A. Therefore, since the motor driver's current is limited to 1A for operation, the effective peak current is 1A.	The NEMA 17 stepper motor has a peak current of 1A; however, the TMC2209 motor driver can supply a peak current of up to 2.8A, enabling it to drive the motor stably even under maximum load [5] [6].

Voltreg_MC_dcpwr: Output

V _{max} = 5.2V	This represents the maximum upper limit for the 5V buck regulator output. A voltage of 5.2V is a safe level for the ESP32-DevKitC.	According to the datasheet, the ESP32-DevKitC v4 can accept an input voltage of up to 7V [11].
V _{min} = 4.8V	This represents the expected minimum output voltage of the	According to the LM2756 datasheet, the fixed 5V output

	buck converter. This value remains stable enough to power the ESP32-DevKitC. The ESP32-DevKitC utilizes its internal regulator to step down the input voltage to a 3.3V logic voltage.	version maintains a tolerance range of within 3% [7].
$I_{nom} = 3A$	The MCU has a peak current of approximately 225 mA. However, the LM2576 is capable of delivering an output of up to 3 A, thereby enabling it to reliably supply power to the MCU.	According to the datasheet, the LM2576 operates as a simple 3A step-down regulator. Therefore, it can continuously supply its rated output current of 3A without overheating [7].

Block 1 Verification Process

Adapter and Voltage Regulator Configuration and Verification:

First, verify the stability of the input interface via the barrel connector. Connect a DC power supply to the board's input terminal, and attach an electronic load to the circuit's main power line to simulate the load of the entire system. Place the probes of a digital multimeter directly onto the input pin of the barrel connector to monitor the voltage supplied by the adapter in real time. The verification process is conducted across four corner cases:

First, test the system for 30 seconds under an input voltage of 12.6V and a load corresponding to the system's nominal current of 1.08A to ensure the voltage remains within the 11.4V to 12.6V range. (Note: The nominal current here represents the sum of the 1A drawn by the two motors and the 80mA drawn by the microcontroller.)

Second, at the same input voltage, increase the load to the system's peak current of 2.25A and verify that the system can sustain this load for 3 seconds.

Third, lower the input voltage to 11.4V and take measurements for 30 seconds under a 1.08A load.

Fourth—and finally—apply the peak load of 2.25A for 3 seconds at an input voltage of 11.4V.

To pass this verification process, the barrel connector and input wiring must maintain a stable voltage range throughout these steps without exhibiting any voltage drop.

Motor Driver Output Configuration and Verification:

Once the input-side verification is complete, inspect the power path leading to the motor driver. Connect the electronic load across the 470 μ F bulk capacitor located at the motor output terminals, and place a digital multimeter in parallel with it. Since the current circuit design does not incorporate an input protection diode, the measured output voltage should be nearly identical to the input voltage. First, with a 12.6V input and the motor operating at its nominal load of 1.0A, conduct a 30-second test to verify that the voltage remains within the range of 11.4V to 12.6V.

Second, with a 12.6V input, apply the motor's peak current—2.0A—for 3 seconds to check for voltage stability.

Third and fourth, lower the input voltage to 11.4V and repeat the same nominal and peak load tests.

In all cases, the voltage must not drop below 11.4V.

Microcontroller Output Configuration and Verification:

Finally, verify the power path for the control section. Connect an electronic load across the 330 μ F capacitor at the output terminal of the LM2576-5.0WT buck converter, and monitor the voltage using a digital multimeter.

First, with a 12.6V input and the control section operating at its nominal load of 80mA, take measurements for 30 seconds to verify that the voltage is precisely maintained within the range of 4.8V to 5.2V.

Second, with a 12.6V input, apply the control section's peak current—250mA—for 3 seconds.

Third and fourth, lower the input voltage to 11.4V and perform the tests under the same nominal and peak load conditions.

Block 1 Artifacts

The NEMA 17 motor has a peak current of 1A, and I decided to integrate two of these motors into my design. Consequently, the instantaneous maximum current—including the MCU—reached approximately 2.25A. To address this, I had to discard the existing voltage regulator and design a new one from scratch. Furthermore, to withstand the high voltage and current levels involved, I had to utilize a wide variety of numerous components.

During this process, I discovered that the specific components I required were not available within the TekBot kit. As a result, I had to manually review datasheets on the Digi-Key website to select the necessary parts—a process that was both time-consuming and extremely arduous. For instance, I needed to design a 5V power line for the microcontroller. I decided to use the LM2576 for this purpose; however, the LM2576 is a buck converter. Since this was my first time working with a buck converter, the concept felt quite unfamiliar to me. Moreover, because a buck converter necessitates the use of a Schottky diode, capacitors, and an inductor in conjunction, the overall complexity of the circuit increased significantly.

Through this experience, I gained firsthand insight into the tangible impact that component availability has on real-world engineering projects.

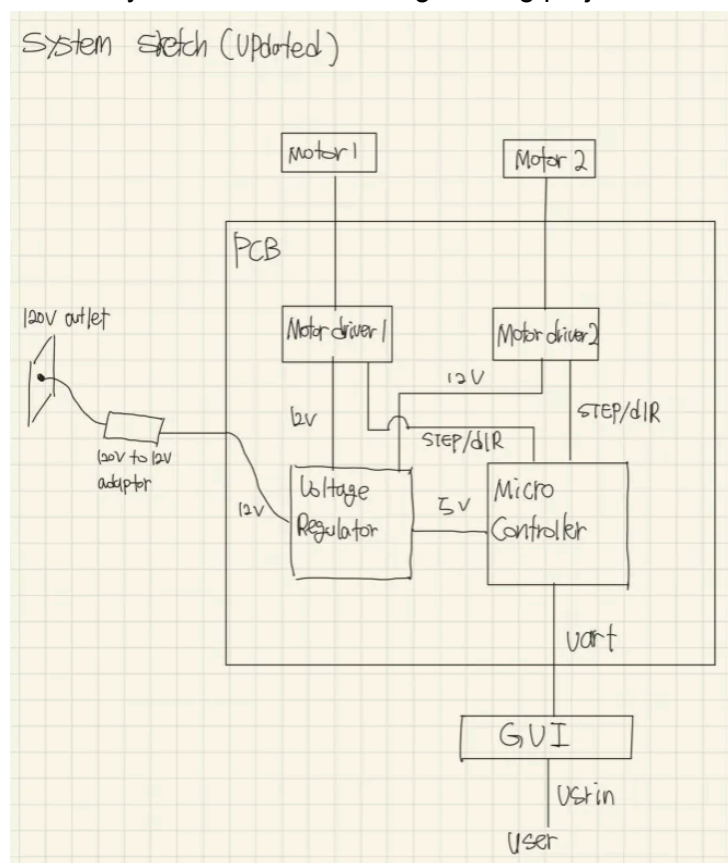


Fig 5. System Sketch.[1]

	Part	Value	Footprint	Supplier	Data sheet	quantity
1	Power Adaptor	12V, 3A	12V DC power, up to 3A	Link	Link	1
2	Barrel Jack	DCJ235-05-A-K1-K	DCJ235-05-A-K1-K_GCT	Link	Link	1
3	Bulk Capacitor (470uF)	UPW1E471MPD	CAP_10 X 16_NCH	Link	Link	1
4	Decoupling Capacitor (100uF)	REF0511101M025K	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm	Link	Link	3
5	Motor Driver	TMC2209 V1.3	Module:Pololu_Breakout-16_15.2x20.3mm	Link	Link	2
6	Motor	Nema 17 Pancake	N/A	Link	Link	2
7	Buck Regulator (5V, 3A)	LM2576-5.0WT	TO-220-5T_MCH	Link	Link	1
8	Output Schottky Diode	1N5822	1N5822	Link	Link	1
9	Output Inductor (330uH)	PE-53117NL	IND_PE-53117NL_PUL	Link	Link	1
10	Output Capacitor (330uF)	EEU-FR1V331UB	CAP_PAN_10X12.5_PAN	Link	Link	1
11	Microcontroller	ESP32-DevKitC V4	MODULE_ESP32-DEVKITC-32D	Link	Link	1

Fig 5. Value and Footprint table.[1]

Block 1 References

[1] Q. Ualtek, "Energy Efficiency Level VI Output Protections: SCP/OCP QFWB-36 Series P P C."

Accessed: Apr. 21, 2026. [Online]. Available:

https://www.qualtekusa.com/images/Power%20Supplies/pdf_files/QFWB-36%20Series.pdf

[2] "PRODUCT SPECIFICATION." Accessed: Apr. 21, 2026. [Online]. Available:

<https://gct.co/files/specs/dcj235-05-spec.pdf>

[3] C. 8100m, "ALUMINUM ELECTROLYTIC CAPACITORS." Available:

https://www.nichicon.co.jp/english/series_items/catalog_pdf/e-upw.pdf

[4] "REF Series FEATURES • Super Low Impedance, High ripple current capability • Endurance: 3000 -6000 hours at 105°C • RoHS Compliance and Halogen-free HOW TO ORDER CASE DIMENSIONS millimeters (inches)." Accessed: Apr. 21, 2026. [Online]. Available:

<https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/8503/REF%20Elect%20Caps%20KAVX.pdf>

[5] "TMC2209 - BIGTREETECH WIKI," *Bttwiki.com*, 2025.

<https://global.bttwiki.com/TMC2209.html>

[6] "E Series Nema 17 Bipolar 1.8deg 17Ncm(24.07oz.in) 1A 42x42x23mm 4 Wires - 17HE08-1004S | StepperOnline," *Omc-stepperonline.com*, 2025.

<https://www.omc-stepperonline.com/e-series-nema-17-bipolar-1-8deg-17ncm-24-07oz-in-1a-42x42x23mm-4-wires-17he08-1004s>

[7] "August 2005 1 LM2576 52kHz Simple 3A Buck Regulator General Description." Accessed: Apr. 21, 2026. [Online]. Available:

<https://ww1.microchip.com/downloads/en/DeviceDoc/lm2576.pdf>

[8] "1N5822 | DigiKey Electronics," *DigiKey Electronics*, 2025.

<https://www.digikey.com/en/products/detail/stmicroelectronics/1N5822/1037332>

[9] PE-53117NL pulse electronics | inductors, coils, chokes | digikey,

<https://www.digikey.com/en/products/detail/pulse-electronics/PE-53117NL/2266021> (accessed Apr. 21, 2026).

[10] EEU-FR1V331U Panasonic electronic components | Aluminum electrolytic capacitors | DigiKey,

<https://www.digikey.com/en/products/detail/panasonic-industry/EEU-FR1V331U/9921026> (accessed Apr. 21, 2026).

[11]“ESP32 Datasheet Espressif Systems,” 2016. Available:
https://cdn.sparkfun.com/datasheets/IoT/esp32_datasheet_en.pdf

Block 1 Future Recommendations

Unlike the voltage regulator design project of the previous semester—for which specific guidelines were provided—this semester required us to start from a completely blank slate. The process of translating basic system requirements into a physically functional and verifiable circuit schematic is extremely challenging; in fact, it remains an ongoing endeavor that I have yet to fully complete.

Nevertheless, I experienced a profound sense of accomplishment after successfully implementing a 5.2V headroom. Through this process, I was able to cultivate a crucial engineering mindset: the ability to anticipate potential edge cases that might arise in real-world applications and to build in sufficient design margins to mitigate them.

If I could offer a piece of advice to my past self—specifically, to the version of me taking the JD1 course last semester—I would tell him this: before diving into the mathematical calculations and design work for a circuit, always verify first whether the necessary components are actually available and compatible with the other parts in the system.

Block 2 Video Link

{Create a record of your block working even if you plan to retest and improve the design. Update this link with the most-recent working verification process. The expectation is you iterate on your designs, but keeping a record will ensure you can show *something* to the instructional team even if everything falls apart at the last minute. This will be your backup verification if something goes wrong or breaks between design, build, and final block testing. Make sure the instructional team has access to view the video.}

Block 2 Description

{Create a block-level block diagram of your individual block. Use IEEE figure labels. Add 1-2 paragraphs describing what this block does. What is its role in the system? How does its role relate to the overall system requirements? What is coming into the block? What does the block do to that input? What is created and then delivered as an output? How? This is where your deep dive into functionality goes. Make sure to include the names and functions of all interfaces related to this block and that they match your top-level architecture above.}

Block 2 Design Details

{Write a 1-2 page detailed description of how this block works. Include at least one schematic or other representative figure. Use IEEE figure labels. Demonstrate your learning by explaining clearly what the inputs are and where they come from. Explain how those inputs become outputs through the functionality of your block. Design details must include in-text citations in IEEE format. Cite resources from prior coursework, module resources lists from this class, or resources you have found externally.}

Block 2 Interface Validation

Interface property *Be sure to use the naming convention "from_to_type" here.	Why is this interface property this value?	How do you know your design details will meet or exceed this property? Cite your sources in IEEE.
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interface_name: input/output

Property 1	{Fill in each section of this table. Include citations, simulation results, calculations, and validation materials in each column that	...
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	address the column prompt above.}	
Property 2	...	...
Property 3	{Make sure you have a minimum of three properties per interface.}	...

interface_name: input/output

Property 1	{Fill in each section of this table. Include citations, simulation results, calculations, and validation materials in each column that address the column prompt above.}	...
Property 2	...	...
Property 3	{Make sure you have a minimum of three properties per interface.}	...

Block 2 Verification Process

1. {Enumerate a verification process here that any junior in the class could follow.
2. Include Setup and Test sections.
3. Be as specific and expository as possible and demonstrate each interface property from the validation table. Explicitly mention when each interface property is verified. What are the pass/fail conditions?
4. Imagine this process will be handed to another team to complete who did not design your system.
5. Write instructions they could follow.}

Block 2 Artifacts

{Populate this section with the miscellaneous but important findings that got you to your final block design. This means anything you had to learn in order to make choices about your design details. This might be prior coursework, examples found online, reference schematics,

pseudocode, previous or prior version block diagrams, etc. Think of this section as a repository of your progress on this block.

If you're wondering what to put in this section, consider what someone looking at your design document would want to see to understand how you got to your final design decisions. This is unique to each block and design engineer.}

Block 2 Future Recommendations

{You have completed the individual technical challenges in this course. Take some time to reflect on how far you have come. How was this different from the Block 1 design cycle? What resources and skills did you utilize to improve during this second design-cycle? What would you tell yourself at the beginning of the term given what you know now? How will you use what you learned in this round in future classes or internships?}

Block 2 References

- [1] Include all relevant IEEE citations here. Be sure to cite them within the text. To get help with this you can search for “IEEE in-text citations” to understand how to cite in-text.
- [2] Cite everything you did not create yourself for this document. This includes but is not limited to diagrams, schematics, pseudocode/code, pinout visuals, etc.